Ardian Mehaj

Junior Software Developer | Python, Backend, Data and Applied AI

Brussels, Belgium | mehajardian@gmail.com

Portfolio | GitHub | LinkedIn

Profile

Computer Science student at OPIT, focused on Python/backend development, data and applied AI. Creator of Synthévia, with two years of work on the project and over a year of personal research into AI trading systems. I define requirements, integrate components and review code and tests, using AI coding assistants for implementation.

Technical skills

Programming : Python (foundations), SQL, TypeScript and JavaScript.

Web and backend : FastAPI, REST APIs, React, HTML/CSS, PostgreSQL and Redis.

Data and machine learning : Data preparation, exploratory analysis, visualisation, supervised classification and model evaluation.

Applied AI : RAG, vector retrieval, prompt engineering, LangChain and LangGraph coursework.

Development workflow : Git, Docker, pytest, Playwright and GitHub Actions; requirements, code review and test review.

Personal projects

Synthévia

Creator and project lead | Personal project | Two years of work

- Conceived and led a platform combining crypto market analysis, trading signals, simulated positions, risk tools and a learning workspace.
- Worked on Python/FastAPI services, React/TypeScript interfaces, PostgreSQL persistence and Redis caching, including account permissions and private content access.
- Researched and reviewed signal scoring and backtests, including data chronology, execution costs, partial exits and evaluation assumptions.
- Integrated a multilingual learning area, server-side assessments and a document-grounded coach with citations and access checks.

synthévia.com

EvidenceDesk

Experimental document review | Python, FastAPI, React, PostgreSQL, Redis

- Worked on passage retrieval, page-linked citations, background processing and per-user access controls in a local prototype.
- Defined checks for answer support and abstention, reviewed implementation and test cases, and documented evaluation limits.

API Contract Guard

Developer tooling | TypeScript, Node.js, OpenAPI

- Scoped a CLI and browser demo comparing five supported categories of API changes, with JSON and HTML reports.
- Defined expected behaviour and reviewed implementation, tests and rejection of unsupported inputs.

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Further projects

ToolCall Replay : Editable trace review with deterministic rules; replay executes no tools or models.

Entity Resolution Workbench : Synthetic record comparison with field-level similarity explanations and a separate Python matcher.

PostgreSQL Migration Rehearsal : A local migration and rollback exercise using a small fixture database.

Skill Studio : Prepared instruction editor with text diffs and Markdown export; the public demo does not run models.

Applied coursework

SpaceX Launch Analysis | IBM Applied Data Science Capstone

Course project covering data preparation, SQL exploration, visualisation, a dashboard and comparison of logistic regression, SVM, decision tree and KNN. Documented preprocessing leakage and evaluation limits; results do not establish future launch performance.

BCG X Data Science Job Simulation | Forage | October 2026

Completed exercises in problem framing, exploratory analysis, data cleaning, feature engineering, modelling and evaluation.

Education

BSc (Hons) Computer Science | OPIT

September 2026 – expected September 2029 | Online programme

Institut Technique René Cartigny | CESS | June 2025 | Belgian upper-secondary diploma, polyvalent mechanics

Certifications

IBM Data Science Professional Certificate
IBM / Coursera | October 2026

IBM RAG and Agentic AI Professional Certificate
IBM / Coursera | October 2026

Google AI Professional Certificate
Google / Coursera | May 2026

HBR Guide to Generative AI for Managers Specialization
Harvard Business Review / Coursera | October 2026

Strengthen Your Business Savvy Specialization
Harvard Business Review / Coursera | October 2026

HackerRank | Python Basic | SQL Advanced

Other experience

Student employee | Delhaize, Brussels | 2020–2022

Coordinated tasks with colleagues and helped organise the team when the manager was absent.

Volunteer tutoring | Several years

Explained mathematics, science and business topics in clear steps adapted to the learner.

Languages and availability

French (main language), Albanian (native), English EF SET 77/100 C2 overall.

EF SET | September 2026

Belgian citizen. Available from January 2027. Open to remote work and relocation; arrangements to be organised.